Hospital Inpatient Web Portal
Detailed Business & Functional Requirements

Hospital-side data entry, care coordination, medication, diagnostics, scheduling, discharge and governance requirements

Prepared as a healthcare software company requirements sheet. This document is structured for product, design, engineering, QA, implementation and hospital stakeholder review. It assumes a secure inpatient workflow where authorized hospital users enter and maintain patient-facing and internal operational data.

Document intent: define the web application requirements for hospital users who create, edit, review and publish inpatient information that later appears in the patient application or portal.

Recommended use: convert this sheet into page wireframes, API contracts, database schema, role matrices, user stories, and QA test cases.

Document Control

Item	Value
Document type	Business Requirements Document (BRD) / Functional Requirements Specification (FRS)
Product scope	Hospital-side inpatient web application
Primary users	Inpatient department staff, nurses, doctors, consultants, diagnostics staff, admin/reception, compliance/admin users
Primary outcome	Capture accurate inpatient information, coordinate care, control patient visibility, and support safe discharge/follow-up workflows
Deployment assumption	Responsive web application for desktop-first hospital operations
Compliance orientation	Ontario/Canada privacy-aware design with PHIPA/PIPEDA considerations; cross-reference HIPAA-style controls where export or U.S. deployments are possible

1. Product Goals and Design Principles

Goal ID	Requirement / Principle	Why it matters
G-01	Single operational source of truth for inpatient stay data	Reduces fragmentation across nursing notes, medications, reports, consultant visits and discharge information.
G-02	Clear separation of internal-only information and patient-visible information	Prevents accidental disclosure of sensitive notes while still supporting patient access to appropriate records.
G-03	Fast, low-friction data entry for hospital staff	Adoption depends on short clicks, clear defaults, and minimal duplicate entry.
G-04	Safe medication, diagnostics and appointment workflows	Time-sensitive clinical operations need validation, alerts and traceability.
G-05	Auditability and accountability for every update	Health software must support privacy, safety, incident review and operational governance.
G-06	Publish-ready patient data model	Hospital-entered data should flow cleanly into the patient app without manual rework.

2. User Roles and Permission Boundaries

Role	Core actions	Restricted actions / notes
Super Admin	Manage hospital tenant, configuration, roles, departments, master data, integrations, policy settings	Should not routinely edit clinical records unless explicitly granted emergency permissions.
Inpatient Coordinator / Ward Staff	Register admissions, maintain patient profile, update care activity, coordinate reports and appointments	Cannot change privileged clinical sign-off fields unless delegated.
Nurse	Update activities, medication administration, vitals/care notes, nursing instructions acknowledgement	Cannot finalize diagnosis or discharge sign-off.
Doctor / Consultant	Enter diagnosis, treatment summary, prescriptions, visit notes, discharge summary approvals	Should not access admin configuration unless separately authorized.
Diagnostics / Lab User	Create test orders, upload reports, change report statuses	Cannot change unrelated medication or discharge content.
Reception / Scheduling	Manage doctor availability, follow-up appointments, reschedule operations	No access to internal clinical notes unless required.

Role	Core actions	Restricted actions / notes
Privacy / Compliance Reviewer	Review access logs, breach investigations, export logs	Read-only or case-based access preferred.

3. Page Map for the Hospital Web Application

Page / Module	Purpose	Primary users
Login & Session Control	Authenticate staff and enforce secure session rules	All hospital users
Dashboard	Operational overview, tasks, alerts and quick navigation	All hospital users
Patient Search / Admission	Create or retrieve patient admission records	Ward staff, reception, coordinators
Patient Record Workspace	Hub with overview, clinical summary, activities, medication, reports and discharge tabs	All authorized clinical users
Medication Workspace	Prescribe, schedule and administer medications	Doctors, nurses
Diagnostics & Reports	Order, schedule, upload and publish reports	Doctors, lab users, ward staff
Scheduling & Follow-up	Consultant visits, follow-up dates and doctor availability	Coordinators, reception, doctors
Discharge Planning	Prepare discharge summary and post-discharge instructions	Doctors, ward staff, nurses
Document Center	Upload and classify care documents	All authorized users
Admin / Audit	User administration, policy settings, access review and logs	Admin, compliance

4. Authentication, Session and Access Control Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
AUTH-01	Secure login	Support username/password and optionally SSO for hospital staff.	Login screen with hospital branding, forgot password, lockout messaging.	Required	Account lockout after configurable failed attempts.
AUTH-02	Role-based access control	Every module, tab, field group and action shall be permission-	System maps users to roles and department scopes.	Required	No user should see patient data outside authorized

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
		driven.			scope.
AUTH-03	Session timeout	Inactive sessions shall auto-expire after a configurable period.	Warn user before session expiry; allow safe save-before-timeout patterns.	Required	Reduces unattended workstation risk.
AUTH-04	Device/session logging	Record login time, IP/device metadata and logout/session termination events.	Accessible to admin/compliance logs.	Required	Supports investigations and suspicious access review.
AUTH-05	Break-glass access	Optional emergency override flow for exceptional access needs.	Require reason capture and extra audit entry.	Optional / Recommended	Useful for urgent care scenarios.

5. Dashboard Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
DB-01	Operational summary cards	Display admitted patients, pending discharge, pending reports, upcoming consultant visits, due medication tasks and unresolved alerts.	Cards should be role-aware and clickable.	Required	Counts must refresh in near real time or on short interval.
DB-02	Task-driven worklist	Show actionable items by role, such as reports to review, medications overdue, appointments needing confirmation.	Filter by unit, doctor, date, status.	Required	Default sort by urgency/time.
DB-03	Global patient search	Search by patient ID, admission ID, name, phone, bed/room.	Search input persistent in header or dashboard.	Required	Return exact and fuzzy matches with unit/ward indicators.
DB-04	Alert center	Prominent alert	Clickable to	Required	Critical alerts

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
	preview	strip or panel for abnormal results, overdue tasks, discharge blockers and unresolved conflicts.	source record.		must be visually differentiated.

6. Patient Search, Registration and Admission Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
ADM-01	Patient identity capture	Form shall capture patient ID/MRN, full legal name, DOB, age, sex/gender, contact information, emergency contact and address.	Admission form with clear required-field indicators.	Required	Prevent incomplete record creation without mandatory fields.
ADM-02	Duplicate detection	System shall warn on likely duplicate patients using MRN, phone, DOB and name combinations.	Present possible matches before creating a new record.	Required	Reduce duplicate patient charts.
ADM-03	Admission details	Capture admission ID, admission date/time, admitting doctor, unit/department, ward, room, bed, initial diagnosis, allergies and status.	Bed/ward selection should come from controlled master data.	Required	Room/bed conflicts blocked.
ADM-04	Status tags	Support inpatient status tags such as stable, observation, critical, isolation, discharge planned.	Status chips visible in search and profile views.	Required	Critical/isolation statuses should trigger visual emphasis.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
ADM-05	Consent and visibility flags	Capture consent acknowledgements, caregiver authorization and patient portal activation state.	Visibility controls grouped in a dedicated panel.	Required	Patient portal remains inactive until required conditions are met.

7. Patient Record Workspace (Overview Tab) Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
OV-01	Single patient header	Persistent header shall show patient name, identifiers, admission status, ward/bed, age, allergies and primary doctor.	Header remains visible or collapsible across tabs.	Required	High-risk alerts such as allergy must always remain visible.
OV-02	Summary cards	Provide quick panels for admission summary, active medications count, pending diagnostics, next appointment and expected discharge date.	Cards link to respective tabs.	Required	Counts must be derived from live child modules.
OV-03	Portal publication status	Indicate which content types are visible to patient app and whether any items are awaiting clinician review before publication.	Publication status shown as badges.	Required	No accidental auto-publication of internal notes.

8. Clinical Summary and Care Notes Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
CLN-01	Diagnosis and treatment	Store provisional diagnosis,	Structured form with long-text	Required	Version history retained for all

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
	summary	confirmed diagnosis, treatment plan, precautions, diet instructions and clinical observations.	areas and sign-off metadata.		edits.
CLN-02	Progress notes	Allow chronological doctor and nursing progress note entries with author and timestamp.	Note composer with role label and optional templates.	Required	Entries immutable after sign-off except via addendum workflow.
CLN-03	Internal vs patient-facing notes	Notes module shall support explicit visibility selection.	Default new notes to internal-only unless role-specific workflow says otherwise.	Required	Patient-facing notes should use simpler language where possible.
CLN-04	Sign-off workflow	Critical summaries may require doctor sign-off and optional nurse acknowledgement before publication or discharge use.	Display sign-off state and blockers.	Recommended	Avoid unsigned discharge content.

9. Daily Activity Timeline Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
ACT-01	Activity logging	Capture date/time, activity type, performed by, status and remarks for inpatient care activities.	Timeline entry form with searchable activity type list.	Required	Timestamp required for every entry.
ACT-02	Activity taxonomy	Support standard activity categories such as doctor round, nursing check,	Master list plus controlled custom additions.	Required	Custom types need admin governance.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
		vitals recorded, medication administered, meal served, physiotherapy, wound care, sample collection, scan completed and patient transfer.			
ACT-03	Timeline visualization	Show entries in chronological timeline with filters by date, author, activity type and status.	Fast-scrolling and sticky date grouping recommended.	Required	Users must locate latest activity quickly.
ACT-04	Escalation markers	Allow priority or exception flags on activities, e.g., missed nursing check or postponed procedure.	Highlighted in timeline and alert center.	Recommended	Critical exceptions must create alerts.

10. Medication Prescribing and Scheduling Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
MED-01	Prescription entry	Capture medicine name, generic name, dose, dosage unit, route, formulation, frequency, purpose, prescribing doctor, start/end dates and instructions.	Medication form with search/autocomplete and structured fields.	Required	Medicine name should map to standardized catalog where possible.
MED-02	Time scheduling	Support exact times, morning/afternoon/evening/night grouping, repeat rules and PRN medications.	Schedule builder with multiple time rows.	Required	At least one valid schedule or PRN rule required.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
MED-03	Food and caution instructions	Capture before food/after food/with food/no restriction plus special warnings.	Dropdown plus free-text note.	Required	Patient-visible instruction must be understandable.
MED-04	Interaction and allergy warnings	System should surface medication-allergy and known interaction warnings where integrated drug data exists.	Non-blocking or blocking alerts based on severity policy.	Recommended	High-severity conflicts require explicit override reason.
MED-05	Change history	Track edits, discontinuation, replacement and pause/resume history.	Medication timeline or audit drawer.	Required	No hard delete for active or historical prescriptions.

11. Medication Administration Log Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
MAR-01	Administration events	Nurses or authorized users shall mark medication as administered, skipped, refused, delayed or unavailable.	Quick action controls from the medication schedule.	Required	Action must capture timestamp and user.
MAR-02	Reason capture	Skipped, delayed or refused medication requires reason text or coded reason.	Prompt appears when exception status selected.	Required	Exception reasons become auditable.
MAR-03	Due and overdue monitoring	System shall flag due, upcoming and overdue administrations.	Color-coded medication worklist.	Required	Overdue events should feed dashboard alerts.

12. Diagnostics, Reports and Result Publication Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
REP-01	Order and schedule tests	Create diagnostic/test record with test name, category, ordering clinician, order date, scheduled date and status.	Order form with category and test selection.	Required	Status should follow controlled workflow.
REP-02	Status workflow	Support statuses such as ordered, scheduled, sample collected, processing, available, reviewed, abnormal/urgent .	Status displayed in timeline/list and filterable.	Required	Status transitions should follow business rules.
REP-03	File upload and preview	Upload PDF/images and preview inside the patient record.	Drag-and-drop plus metadata panel.	Required	Accepted file types and size limits must be enforced.
REP-04	Doctor review before patient visibility	Support reviewed-by-clinician state before release to patient portal for sensitive result types.	Review checkbox/status with reviewer name and time.	Required	Policy-configurable by report category.
REP-05	Abnormal result flagging	Abnormal or urgent results should trigger visible flags and notifications.	Use badges and alert routing.	Required	Critical values should not rely on passive viewing alone.

13. Consultant Visit, Doctor Availability and Follow-up Scheduling Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
SCH-01	Consult request management	Capture consultant name, specialty, requested date/time, reason, status	Consult request form plus status tracker.	Required	Status options: requested, confirmed, completed, postponed, cancelled.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
		and post-visit notes.			
SCH-02	Doctor availability slots	Maintain doctor calendar with next 4-5 available dates and time slots at minimum.	Calendar/grid view with slot duration and location.	Required	Unavailable dates and blocked slots must be supported.
SCH-03	Follow-up booking	Schedule review visits, repeat tests or post-discharge appointments from inside the patient record.	One-click create from discharge/follow-up panel.	Required	Avoid duplicate bookings for same date/time/provider.
SCH-04	Reschedule/ cancel workflow	Authorized users can reschedule or cancel with reason capture.	Reason visible in audit trail and patient notifications as appropriate.	Recommended	User must see affected patient communications.

14. Discharge Planning and Patient Handover Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
DIS-01	Discharge readiness workspace	Track expected discharge date, blockers, counseling status and discharge checklist completion.	Checklist-style panel with owner and due status.	Required	Discharge should not finalize if mandatory fields are missing.
DIS-02	Discharge summary generation	Capture final diagnosis, hospital course summary, procedures, discharge medications, home instructions, follow-up appointments and warning signs.	Structured summary form with print/export view.	Required	Finalizing should require authorized clinician sign-off.
DIS-03	Patient handoff	Publish	Publication	Required	Only approved

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
	publishing	approved discharge pack to patient app/portal.	status badge and send/notify action.		content becomes patient visible.
DIS-04	Post-discharge reminders	Configure reminder dates for medications, doctor review and scheduled tests.	Reminder setup integrated with follow-up panel.	Recommended	Supports patient-side engagement without duplicate entry.

15. Document Center and File Governance Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
DOC-01	Document repository	Store prescriptions, reports, consent forms, discharge files, referrals and scanned documents.	Document center with category filters and previews.	Required	Every document needs owner, patient linkage and upload timestamp.
DOC-02	Classification and visibility	Each file shall have category, document date, source and patient-visibility setting.	Metadata side panel required on upload.	Required	Patient-visible defaults should be conservative.
DOC-03	Versioning / replacement	Allow updated files while keeping prior versions for traceability.	Show current vs superseded files.	Recommended	No destructive overwrite without history.

16. Notifications, Alerts and Communication Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
NOT-01	Operational alerts	Generate alerts for overdue medication, pending abnormal results, same-day consultant visits,	Notification center plus dashboard surfacing.	Required	Critical alerts need severity levels.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
		discharge blockers and missed activities.			
NOT-02	User-directed reminders	Assign reminders/tasks to staff users or teams.	Task owner, due date and status controls.	Recommended	Helps close operational gaps.
NOT-03	Patient communication triggers	Hospital actions such as report release, appointment booking or discharge publication should be able to trigger patient app notifications.	Event hooks configurable by content type.	Recommended	Message templates must avoid exposing unpublished data.

17. Admin, Audit and Governance Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
AUD-01	Audit trail	Record create, edit, status change, sign-off, publish and download events with user, timestamp and affected field/object.	Audit viewer with filters and export.	Required	Audit data should be tamper-evident and retained per policy.
AUD-02	Role and master data administration	Manage users, departments, activity types, report categories, doctor masters, beds/rooms and schedule templates.	Admin console with import/export tools if needed.	Required	Changes to master data should be separately audited.
AUD-03	Access review support	Allow compliance/privacy users to review who accessed specific patient records.	Patient-centric access log view.	Required	Essential for privacy investigations.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
AUD-04	Data export controls	Limit bulk export and require permission for sensitive data extracts.	Export actions logged with purpose where possible.	Recommended	Prevents uncontrolled data leakage.

18. Non-Functional, Security and Compliance-Oriented Requirements

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
NFR-01	Availability and performance	Patient search and patient record load should be fast enough for point-of-care use; core screens should remain responsive under concurrent ward usage.	Target metrics to be finalized in architecture phase.	Required	Performance budgets needed before build.
NFR-02	Encryption and secure transport	Use strong transport encryption and protect stored sensitive data with appropriate safeguards.	No plaintext PHI in logs or downloads.	Required	Align with organizational security standards.
NFR-03	Least privilege and segregation	Default access should be limited by role, department and care relationship where appropriate.	Permission model must be configurable.	Required	Supports PHI minimization.
NFR-04	Retention, backup and recovery	Support backup, restore, retention schedules and disaster recovery procedures.	Operational runbooks required.	Required	Health records are business-critical.
NFR-05	Accessibility and usability	Use readable forms, clear labels, keyboard support and high-contrast	Design system should follow accessibility best practices.	Recommended	Helps diverse staff populations and reduces entry errors.

Req ID	Feature	Requirement detail	UI / Workflow expectations	Priority	Validation / Rules
		status indicators for clinical environments.			
NFR-06	Traceable patient publication model	Internal data and patient-published data states must be explicit and reviewable.	Content state machine recommended: draft, reviewed, published, withdrawn.	Required	Prevents accidental release of incomplete data.

19. Detailed Input Field Catalogue by Webpage Segment

19.1 Admission Form Fields

Field group	Fields	Mandatory subset	Validation / notes
Demographics	Patient ID/MRN; full name; DOB; age; sex/gender; phone; address; emergency contact; relationship; blood group	Patient ID, full name, DOB, admission date/time, emergency contact	Phone format validation; DOB cannot be future date; age derived where possible
Admission details	Admission ID; date/time; department; ward; room; bed; admitting doctor; initial diagnosis; allergies; status tags	Admission ID, date/time, department, bed/room, admitting doctor	Room/bed conflicts blocked; allergy field always visible
Portal/consent	Portal activation status; caregiver authorized; consent timestamps; language preference	Portal status if patient app is enabled	Patient visibility should remain disabled until prerequisites complete

19.2 Clinical Summary Fields

Field group	Fields	Mandatory subset	Validation / notes
Diagnosis & plan	Provisional diagnosis; final diagnosis; treatment plan; precautions; diet instructions; patient condition summary	At least provisional diagnosis and treatment plan	Long-text areas need autosave and version history
Notes metadata	Author; role; timestamp; sign-off state; visibility flag	Author, timestamp, visibility	Signed notes become addendum-only

19.3 Medication Fields

Field group	Fields	Mandatory subset	Validation / notes
Prescription	Drug name; generic name; dose; unit; route; formulation; frequency; purpose; prescriber; instructions	Drug name, dose, route, frequency, prescriber	Use standardized medicine source where available
Schedule	Exact times; days; PRN flag; start date; end date; before/after food; caution text	At least one schedule rule or PRN flag	End date cannot precede start date
Administration log	Administration status; time given; administered by; exception reason	Status and timestamp	Skipped/refused/delayed requires reason

19.4 Diagnostics & Reports Fields

Field group	Fields	Mandatory subset	Validation / notes
Test order	Test/report name; category; ordered by; order date; scheduled date; status	Name, category, order date, status	Controlled status transitions recommended
Result file	Upload file; result date; reviewed by; abnormal flag; patient visibility; notes	File, result date or status depending on workflow	Review-before-release policy configurable

19.5 Scheduling & Follow-up Fields

Field group	Fields	Mandatory subset	Validation / notes
Consult visit	Consultant name; specialty; date; time; reason; status; notes	Consultant, date, time/status	No overlapping provider slots
Doctor availability	Provider; clinic/location; available dates; slot duration; max patient count; blocked dates	Provider, dates, slots	At least 4-5 upcoming dates should be manageable
Follow-up	Review appointment; next test date; teleconsultation/clinic type; reminders	Follow-up date and provider/service	Duplicate follow-up checks recommended

19.6 Discharge & Handover Fields

Field group	Fields	Mandatory subset	Validation / notes
Discharge summary	Final diagnosis; hospital course; procedures; discharge date; medications; home instructions; warning signs; follow-up appointments	Final diagnosis, discharge date, medications/instructions	Finalization requires authorized sign-off
Checklist	Counseling complete; documents shared; reminders enabled; caregiver informed	Required items per hospital policy	Use checklist gating before publish

20. Business Rules to Confirm During Discovery

- Which report categories require clinician review before becoming visible to patients?
- Which roles can create patient-facing notes versus internal-only notes?
- What is the target lockout/session timeout policy for shared clinical workstations?
- Which medication data source or formulary will be used for drug search, warnings and coding?
- How should caregiver access and consent be captured for minors, seniors and dependent adults?
- What discharge checklist items are mandatory per hospital department or care pathway?
- What notification channels are in scope for the patient side: in-app only, SMS, email, push, or combinations?
- Will the product integrate with existing HIS/EHR/LIS/PACS/appointment systems, or operate as a standalone portal initially?

21. Product Feedback and Implementation Advice

Area	Recommendation
Workflow design	Use a persistent patient workspace with tabbed navigation rather than separate disconnected forms. This reduces context switching and makes the record feel like one coherent chart.
Patient publication model	Treat patient visibility as a deliberate publishing workflow, not a side effect of staff editing. Add review states and badges.
Medication safety	Prioritize medication schedule builder, overdue monitoring and exception reason capture in MVP. These are high-value operational workflows.
Diagnostics release policy	Make report release rules configurable by report category and organization policy. Some tests should require review before release.
Search and speed	Invest early in very fast patient lookup, sticky patient header, recent records and autosave. Slow workflows will hurt adoption.
Governance	Do not postpone audit logging, field history and access

Area	Recommendation
	traceability to a later phase. These are foundational requirements, not polish features.

Research-Informed Notes

This requirements sheet emphasizes role-based access, access logging, safeguards, patient access governance and controlled disclosure because those patterns are strongly reinforced by official privacy and health IT guidance for health information handling and patient record access.